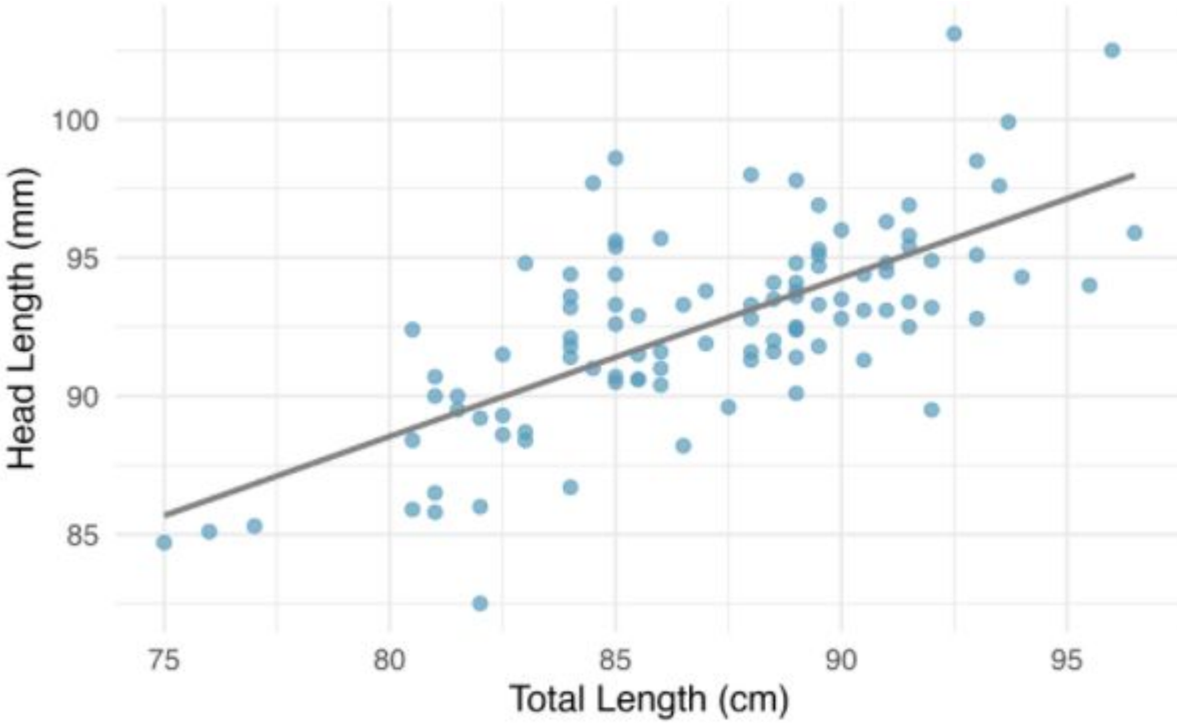
Predicted values, residuals, intercepts, slopes

MATH 213

Linear regression

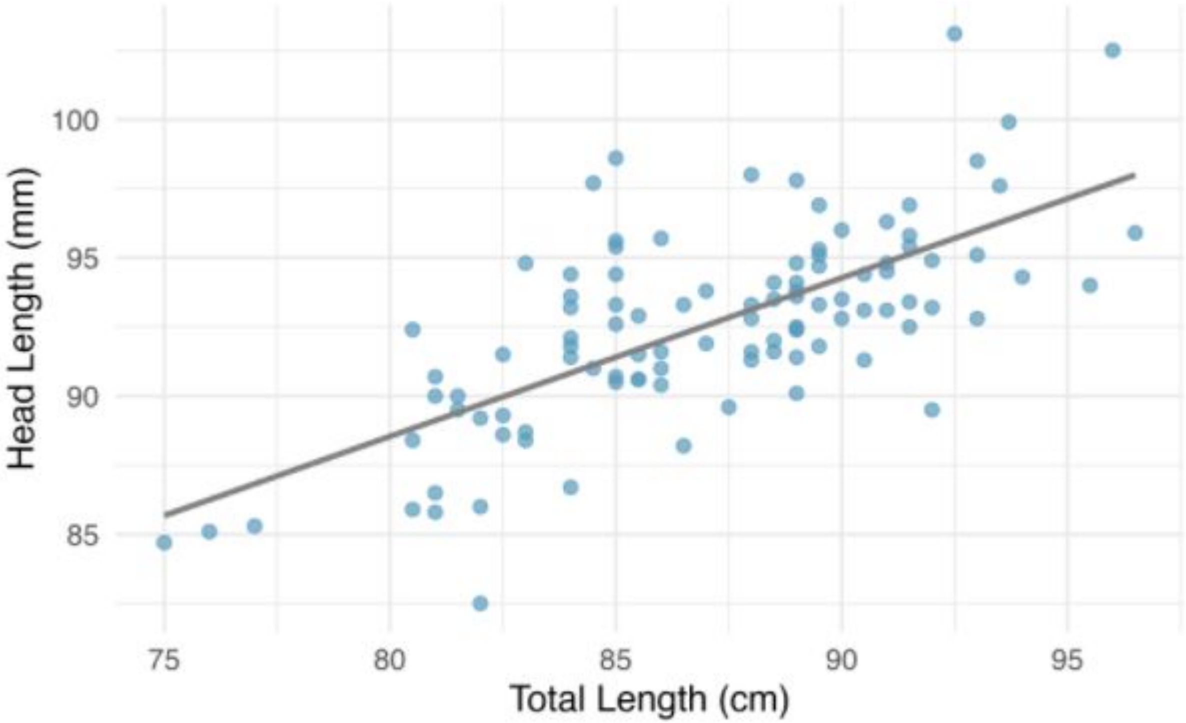


Equation of this regression line:

$$\hat{y} = 41 + 0.59x$$

Calculate the predicted head length for a possum with total length 80cm.

Linear regression



Equation of this regression line:

$$\hat{y} = 41 + 0.59x$$

Calculate the predicted head length for a possum with total length 80cm.

$$\hat{y} = 41 + 0.59 \times 80 = 88.2$$

The predicted head length for a possum of total length 80cm is 88.2mm

Interpretation of slope and intercept

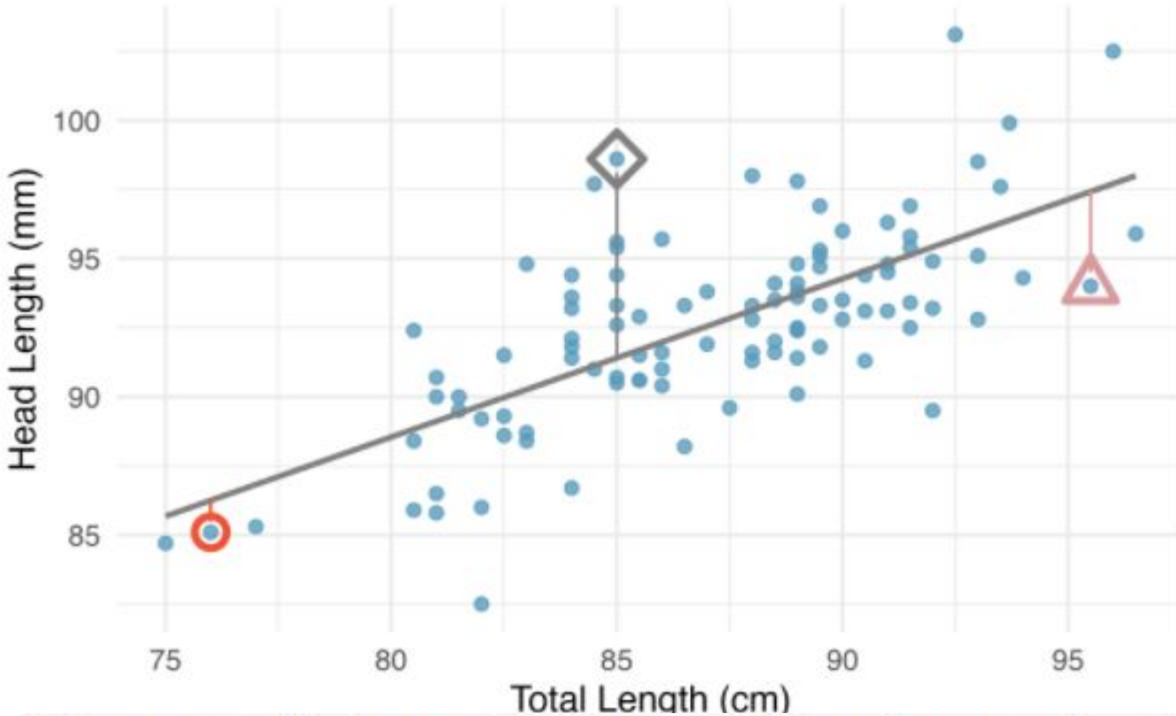
Again, the predicted head length is given by:

$$\hat{y} = 41 + 0.59x$$

Interpret the intercept: The intercept of 41mm is the value of the predicted head length for a total body length of 0 cm. This interpretation makes no sense in this context because there is no such thing as a possum with 0cm body length!

Interpret the slope: For each additional centimeter of total body length, the predicted length of a possum's head increases by 0.59mm.

Residuals

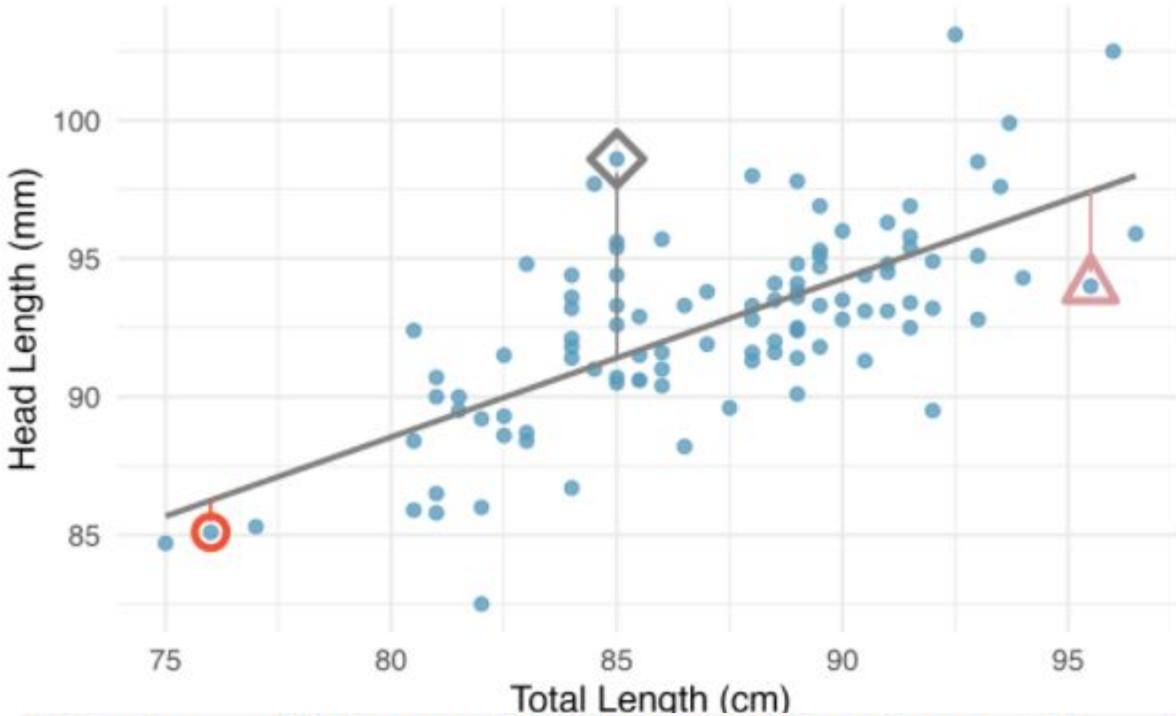


The residual of a prediction is the difference between the observed value and the predicted value.

$$e_i = y_i - \hat{y}_i$$

The linear fit shown in Figure 7.8 is given as $\hat{y} = 41 + 0.59x$. Based on this line, formally compute the residual of the observation $(76.0, 85.1)$. This observation is marked by a red circle in Figure 7.8. Check it against the earlier visual estimate, -1.

Residuals



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$$\hat{y} = 41 + 0.59x = 41 + 0.59 \times 76.0 = 85.84$$

$$e = y - \hat{y} = 85.1 - 85.84 = -0.74$$

Residual is -0.74mm.